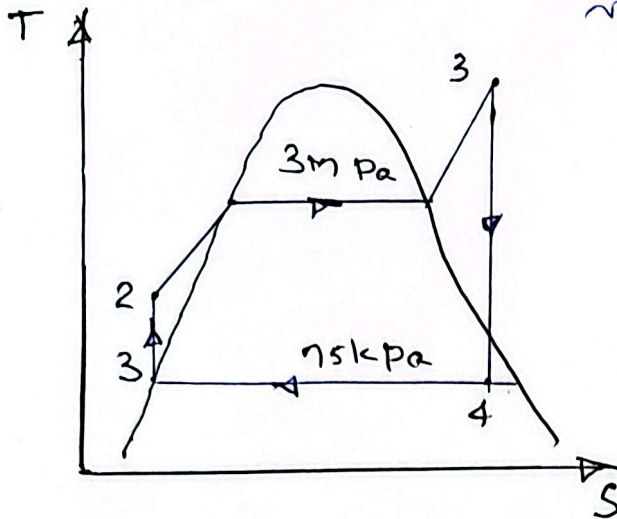


Example 1 The Simple Ideal Rankine cycle $\eta_{th} = ?$



Enthalpies at various point, using data from steam tables (Tables A-4, A-5, and A-6):

$$\text{State 1: } \left. \begin{array}{l} P_1 = 75 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right\} \begin{array}{l} h_1 = h_f @ 75 \text{ kPa} = 384.44 \text{ kJ/kg} \\ v_1 = v_f @ 75 \text{ kPa} = 0.001037 \text{ m}^3/\text{kg} \end{array}$$

$$\text{State 2: } P_2 = 3 \text{ MPa}$$

$$S_2 = S_1$$

$$W_{\text{pump, in}} = v_1 (P_2 - P_1) = (0.001037) (3000 - 75) \text{ kPa} \cdot \left(\frac{\text{kPa}}{\text{kgm}} \right) = 3.03 \text{ kJ/kg}$$

$$h_2 = h_1 + W_{\text{pump, in}} = (384.44 + 3.03) \text{ kJ/kg} = 387.47 \text{ kJ/kg}$$

$$\text{State 3: } \left. \begin{array}{l} P_3 = 3 \text{ MPa} \\ T_3 = 350^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3116.1 \text{ kJ/kg} \\ S_3 = 6.7450 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\text{State 4: } P_4 = 75 \text{ kPa (Sat. mixture)}$$

$$S_4 = S_3$$

$$x_4 = \frac{S_4 - S_f}{S_{fg}} = \frac{6.7450 - 1.2132}{6.2426} = 0.8861$$

$$h_4 = h_f + x_4 h_{fg} = 384.4 + 0.8861 (2278) = 2403 \text{ kJ/kg}$$

Thus,

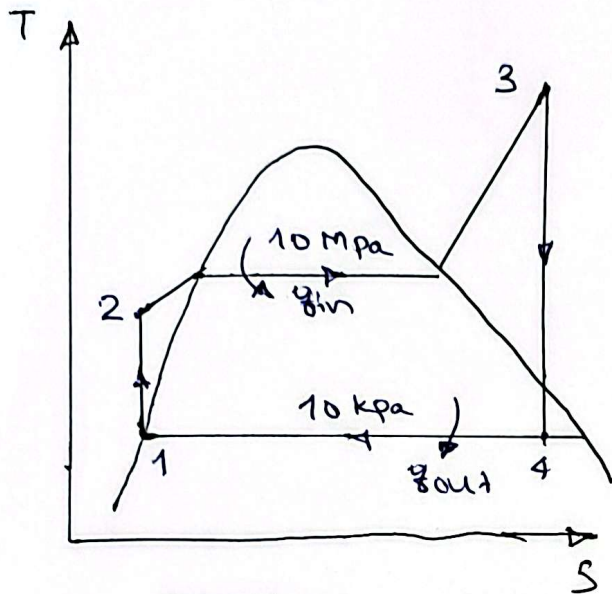
$$q_{\text{in}} = h_3 - h_2 = (3116.1 - 387.47) \text{ kJ/kg} = 2728.6 \text{ kJ/kg}$$

$$q_{\text{out}} = h_4 - h_1 = (2403 - 384.44) \text{ kJ/kg} = 2018.6 \text{ kJ/kg}$$

$$\eta_{th} = 1 - \frac{q_{\text{out}}}{q_{\text{in}}} = 1 - \frac{2018.6 \text{ kJ/kg}}{2728.6 \text{ kJ/kg}} = 0.26 \text{ or } 26\%$$

$$\text{Or } \eta_{th} = \frac{W_{\text{net}}}{q_{\text{in}}} = \frac{q_{\text{in}} - q_{\text{out}}}{q_{\text{in}}} = \frac{710 \text{ kJ/kg}}{2728.6} = 26\%$$

Example 2



$$x_4 = ?$$

$$\eta_{th} = ?$$

$$\dot{m} = ?$$

Solution

a) Use steam tables to find enthalpies at various points.

$$\text{State 1: } P_1 = 10 \text{ kPa} \left\{ \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg} \\ \text{Sat. liquid} \end{array} \right. \left\{ \begin{array}{l} v_1 = v_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg} \end{array} \right.$$

$$\text{State 2: } w_{p,in} = v_1(P_2 - P_1)$$

$$= (0.00101 \text{ m}^3/\text{kg})(10,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kPa}}{1 \text{ kPa} \cdot \text{m}^3} \right)$$

$$= 10.09 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{p,in} = 191.81 + 10.09 = 201.90 \text{ kJ/kg}$$

$$\text{State 3: } P_3 = 10 \text{ MPa} \left\{ \begin{array}{l} h_3 = 3375.1 \text{ kJ/kg} \\ T_3 = 500^\circ\text{C} \end{array} \right. \left\{ \begin{array}{l} s_3 = 6.5995 \text{ kJ/kg} \cdot \text{K} \end{array} \right.$$

$$\text{State 4: } P_4 = 10 \text{ kPa} \left\{ \begin{array}{l} x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.5995 - 0.6492}{7.4996} \\ s_4 = s_3 \end{array} \right. = 0.7934$$

$$h_4 = h_f + x_4 h_{fg}$$

$$= 191.81 + (0.7934)(2392.1) = 2089.7$$

$$b) \dot{q}_{in} = h_3 - h_2 = 3375.1 - 201.9 = 3173.2 \text{ kJ/kg}$$

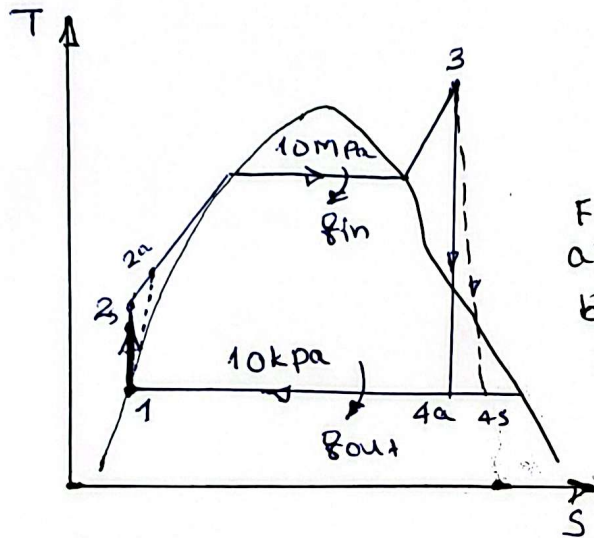
$$\dot{q}_{out} = h_4 - h_1 = 2089.7 - 191.81 = 1897.9 \text{ kJ/kg}$$

$$w_{net} = \dot{q}_{in} - \dot{q}_{out} = 3173.2 - 1897.9 = 1275.4 \text{ kJ/kg}$$

$$\eta_{th} = \frac{w_{net}}{\dot{q}_{out}} = \frac{1275.4 \text{ kJ/kg}}{3173.2 \text{ kJ/kg}} = 40\%$$

$$c) \dot{m} = \frac{\dot{w}_{net}}{w_{net}} = \frac{210,000 \text{ kJ/s}}{1275.4 \text{ kJ/kg}} = 164.7 \text{ kg/s}$$

Example 3



$$\eta_T = 85\%$$

$$\eta_P = 85\%$$

Find
a) Quality

b) η_{th}

c) $\dot{m}$

Solution

a) From steam tables (Tables A-4, A-5 & A-6)
State 1: $P_1 = 10 \text{ kPa}$ } $h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg}$
Sat. liquid } $v_1 = v_f @ 10 \text{ kPa} = 0.0010 \text{ m}^3/\text{kg}$

$$\text{State 2: } w_{P,in} = v_1 (P_2 - P_1) / \eta_P$$

$$= (0.00101 \text{ m}^3/\text{kg}) (10,000 - 10 \text{ kPa}) \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right) / 0.85$$

$$= 11.87 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{P,in} = 191.81 + 11.87 = 203.68 \text{ kJ/kg}$$

State 3: $P_3 = 10 \text{ MPa}$ } $h_3 = 3375.1 \text{ kJ/kg}$
 $T_3 = 500^\circ\text{C}$ } $s_3 = 6.5995 \text{ kJ/kg} \cdot \text{K}$

$$\text{State 4: } P_4 = 10 \text{ kPa} \left\{ \begin{array}{l} x_{4s} = \frac{s_{4s} - s_f}{s_{fg}} = \frac{6.5995 - 0.6492}{7.4996} \\ s_{4s} = s_3 \end{array} \right.$$

$$= 0.7934$$

$$h_{4s} = h_f + x_{4s} h_{fg} = 191.81 + (0.7934)(2392.1) = 2089.7 \text{ kJ/kg}$$

$$\eta_T = \frac{h_3 - h_{4a}}{h_3 - h_{4s}} \rightarrow h_{4a} = h_3 - \eta_T (h_3 - h_{4s})$$

$$h_{4a} = 3375.1 - (0.85)(3375.1 - 2089.7)$$

$$= 2282.5 \text{ kJ/kg}$$

$$\left. \begin{array}{l} P_4 = 10 \text{ kPa} \\ h_{4a} = 2282.5 \end{array} \right\} x_{4a} = \frac{h_{4a} - h_f}{h_{fg}} = \frac{2282.5 - 191.81}{2392.1} = 0.874$$

$$b) \dot{q}_{in} = h_3 - h_2 = 3375.1 - 203.68 = 3171.4 \text{ kJ/kg}$$

$$\dot{q}_{out} = h_4 - h_1 = 2282.5 - 191.81 = 2090.7 \text{ kJ/kg}$$

$$W_{net} = \dot{q}_{in} - \dot{q}_{out} = 3171.4 - 2090.7 = 1080.7 \text{ kJ/kg}$$

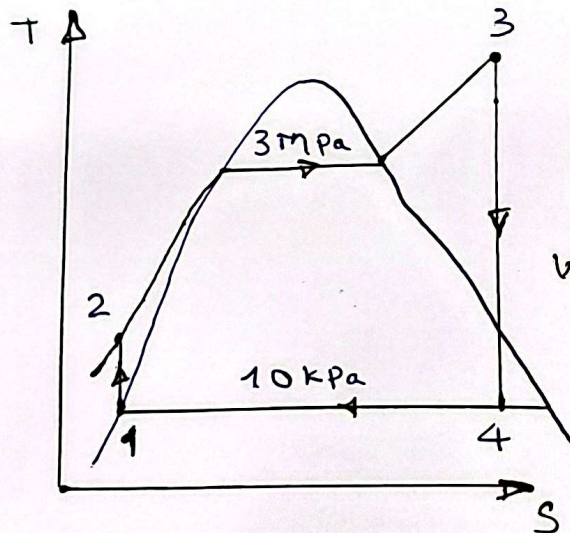
$$\eta_{th} = \frac{W_{net}}{\dot{q}_{in}} = \frac{1080.7 \text{ kJ/kg}}{3171.5 \text{ kJ/kg}} = 34.1\%$$

$$c) \dot{m} = \frac{\dot{W}_{net}}{W_{net}} = \frac{210,000 \text{ kJ/s}}{1080.7 \text{ kJ/kg}} = 194.3 \text{ kg/s}$$

EXAMPLE 4a) This is the steam power plant discussed in example 1.

$$P_1 = 10 \text{ kPa}, P_2 = P_3 = 3 \text{ MPa}, T_3 = 350^\circ\text{C} : \eta_{th} = ?$$

Solution



$$\text{State 1: } P_1 = 10 \text{ kPa} \left. \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right\} \begin{array}{l} = 191.8 \text{ kJ/kg} \\ v_1 = v_f @ 10 \text{ kPa} \end{array}$$

$$v_1 = 0.00101 \text{ m}^3/\text{kg}$$

$$\text{State 2: } P_2 = 3 \text{ MPa} \\ s_2 = s_1$$

$$W_{p,in} = v_1 (P_2 - P_1) \\ = (0.00101 \text{ m}^3/\text{kg}) [(3000 - 10) \text{ kPa}] \\ = 3.02 \text{ kJ/kg} \left(\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right)$$

$$h_2 = h_1 + W_{p,in} = (191.81 + 3.02) \text{ kJ/kg} = 194.83 \text{ kJ/kg}$$

$$\text{State 3: } P_3 = 3 \text{ MPa} \left. \begin{array}{l} h_3 = 3116.1 \text{ kJ/kg} \\ T_3 = 350^\circ\text{C} \end{array} \right\} s_3 = 6.7450 \text{ kJ/kg} \cdot \text{K}$$

$$\text{State 4: } P_4 = 10 \text{ kPa (Sat. mixture)}$$

$$s_4 = s_3$$

$$x_4 = \frac{s_4 - s_f}{s_{fg}} = \frac{6.7450 - 0.6192}{7.4996} = 0.8128$$

$$h_4 = h_f + x_4 h_{fg} = 191.81 + 0.8128 (2392.1) = 2136.1 \text{ kJ/kg}$$

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$$q_{in} = h_3 - h_2 = (3116.1 - 194.83) \text{ kJ/kg} = 2921.3 \text{ kJ/kg}$$

$$q_{out} = h_4 - h_1 = (2136.1 - 191.81) \text{ kJ/kg} = 1944.3 \text{ kJ/kg}$$

Q

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1944.3 \text{ kJ/kg}}{2921.3 \text{ kJ/kg}} = 33.4\%$$

As a result of lowering the condenser pressure η_{th} increases from 26% to 33.4%. P from 75 kPa to 10 kPa.

4b) $P_3 = P_2 = 3 \text{ MPa}$, $T_3 = 600^\circ\text{C}$, state 4 (10 kPa & $s_4 = s_3$) are determined.

States 1 & 2 remain the same as 4a in this case.

$$h_3 = 3682.8 \text{ kJ/kg}$$

$$h_4 = 2380.3 \text{ kJ/kg} \quad (x_4 = 0.915)$$

Thus,

$$q_{in} = h_3 - h_2 = 3682.8 - 194.83 = 3488 \text{ kJ/kg}$$

$$q_{out} = h_4 - h_1 = 2380.3 - 191.81 = 2188.5 \text{ kJ/kg}$$

Q

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{2188.5 \text{ kJ/kg}}{3488 \text{ kJ/kg}} = 0.373 \text{ or } 37.3\%$$

Thermal efficiency increases from 33.4 to 37.3 percent as a result of superheating the steam from 350 to 600°C . At the same time, the quality of the steam increases from 81.3 to 91.5 percent. moisture content decreases from 18.7 to 8.5 percent.

C) State 1 remains the same in this case.

$$h_2 = 206.95 \text{ kJ/kg}$$

$$h_3 = 3583.1 \text{ kJ/kg}$$

$$h_4 = 2115.3 \text{ kJ/kg} \quad (x_4 = 0.804)$$

Thus,

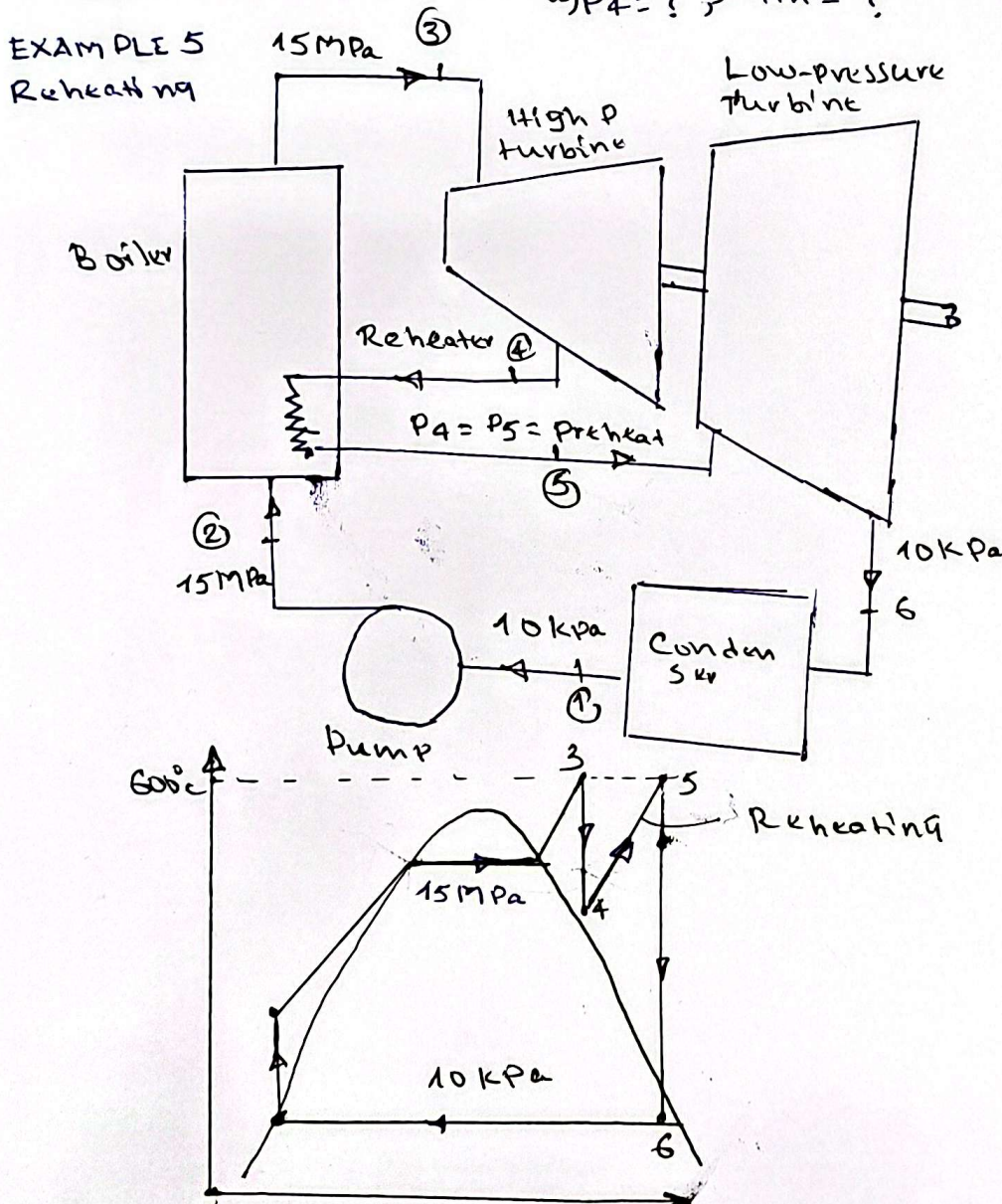
$$q_{in} = h_3 - h_2 = 3583.1 - 206.95 = 3376.2 \text{ kJ/kg}$$

$$q_{out} = h_4 - h_1 = 2115.3 - 191.81 = 1923.5 \text{ kJ/kg}$$

$$\eta_{th} = 1 - \frac{q_{out}}{q_{in}} = 1 - \frac{1923.5 \text{ kJ/kg}}{3376.2 \text{ kJ/kg}} = 0.43 \text{ or } 43\%$$

The thermal efficiency increases from 37.3 to 43 Percent as a result of raising the boiler pressure from 3 to 15 MPa while maintaining the turbine inlet temperature to 600°C. At the same time, however the quality of the steam decreases from 91.5 to 80.4 Percent (or the moisture content increases from 8.5 to 19.6 Percent)

a) $P_4 = ?$ b) $\eta_{th} = ?$



a) The reheat pressure is determined from the requirement that the entropies at states 5 & 6 be the same.
State 6: $P_6 = 10 \text{ kPa}$

$$x_6 = 0.896 \text{ (Sat. mixture)}$$

$$s_6 = s_f + x_6 s_{fg} = 0.6492 + (0.896)(7.4996) = 7.3638$$

$$h_6 = h_f + x_6 h_{fg} = 191.81 + 0.896(2392.1) = 2335.1 \text{ kJ/kg}$$

$$\text{State 5: } \left. \begin{array}{l} T_5 = 600^\circ\text{C} \\ S_5 = S_6 \end{array} \right\} \begin{array}{l} P_5 = 4 \text{ MPa} \\ h_5 = 3674.9 \text{ kJ/kg} \end{array}$$

Therefore, Steam should be reheated at a pressure of 4 MPa or lower to prevent a moisture content above 10.4%.

b) To determine the thermal efficiency, we need to know the enthalpies at all states:

$$\text{State 1: } \left. \begin{array}{l} P_1 = 10 \text{ kPa} \\ \text{Sat. liquid} \end{array} \right\} \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} = 191.81 \text{ kJ/kg} \\ v_1 = v_f @ 10 \text{ kPa} = 0.00101 \text{ m}^3/\text{kg} \end{array}$$

$$\text{State 2: } P_2 = 15 \text{ MPa}$$

$$S_2 = S_1$$

$$w_{p,in} = v_1 (P_2 - P_1) = (0.00101 \text{ m}^3/\text{kg}) ([15,000 - 10] \text{ kPa}) \left[\frac{1 \text{ kJ}}{1 \text{ kPa} \cdot \text{m}^3} \right]$$

$$= 15.14 \text{ kJ/kg}$$

$$h_2 = h_1 + w_{p,in} = (191.81 + 15.14) \text{ kJ/kg} = 206.95 \text{ kJ/kg}$$

$$\text{State 3: } \left. \begin{array}{l} P_3 = 15 \text{ MPa} \\ T_3 = 600^\circ\text{C} \end{array} \right\} \begin{array}{l} h_3 = 3583.1 \text{ kJ/kg} \\ S_3 = 6.6796 \text{ kJ/kg} \end{array}$$

$$\text{State 4: } \left. \begin{array}{l} P_4 = 4 \text{ MPa} \\ S_4 = S_3 \end{array} \right\} \begin{array}{l} h_4 = 3155 \text{ kJ/kg} \\ (T_4 = 375.5^\circ\text{C}) \end{array}$$

• Thus,

$$\dot{q}_{in} = (h_3 - h_2) + (h_5 - h_4)$$

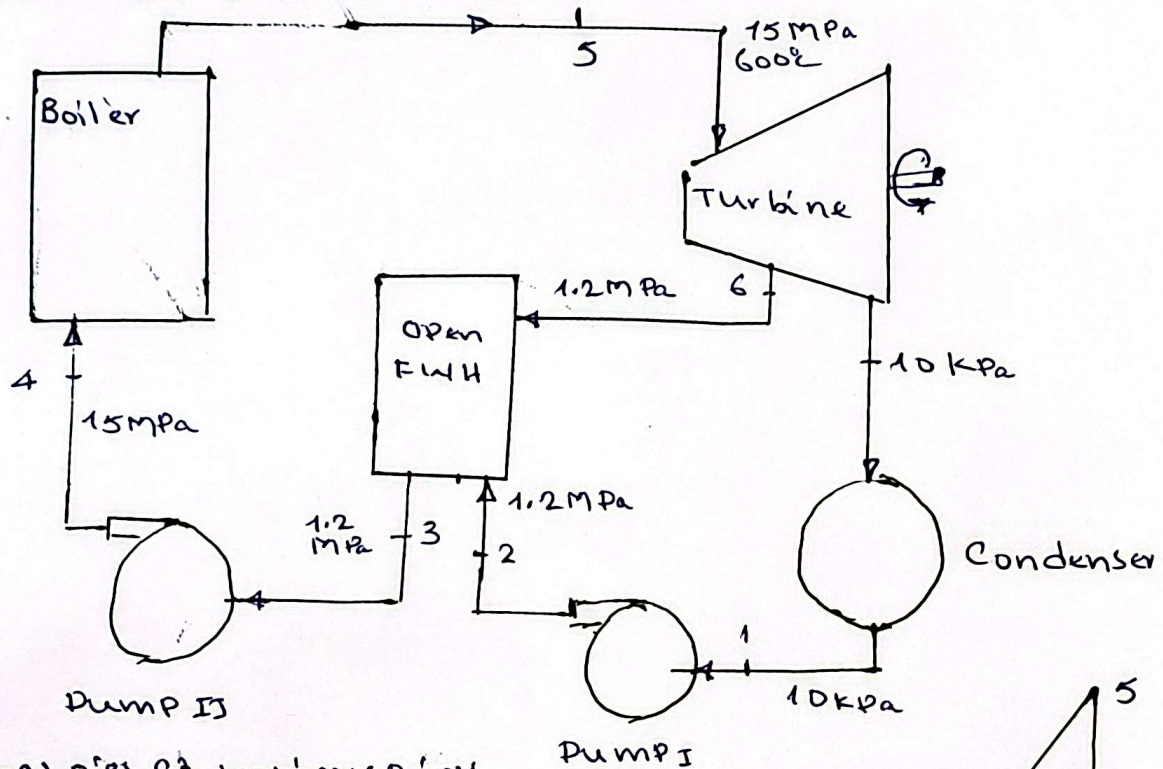
$$= (3583.1 - 206.95) \text{ kJ/kg} + (3674.9 - 3155) \text{ kJ/kg}$$

$$= 3896.1 \text{ kJ/kg}$$

$$\dot{q}_{out} = h_6 - h_1 = (2335.1 - 191.81) \text{ kJ/kg} = 2143.3 \text{ kJ/kg}$$

$$\eta_{th} = 1 - \frac{\dot{q}_{out}}{\dot{q}_{in}} = 1 - \frac{2143.3 \text{ kJ/kg}}{3896.1 \text{ kJ/kg}} = 0.45 \text{ or } 45\%$$

EXAMPLE 6 (Ideal Regenerative Rankine cycle)



Enthalpies at various points

$$\text{State 1: } P_1 = 10 \text{ kPa} \left\{ \begin{array}{l} h_1 = h_f @ 10 \text{ kPa} \\ = 191.81 \\ v_1 = v_f @ 10 \text{ kPa} \\ = 0.00101 \text{ m}^3/\text{kg} \end{array} \right.$$

$$\text{State 2: } P_2 = 1.2 \text{ MPa} \left\{ \begin{array}{l} s_2 = s_1 \end{array} \right.$$

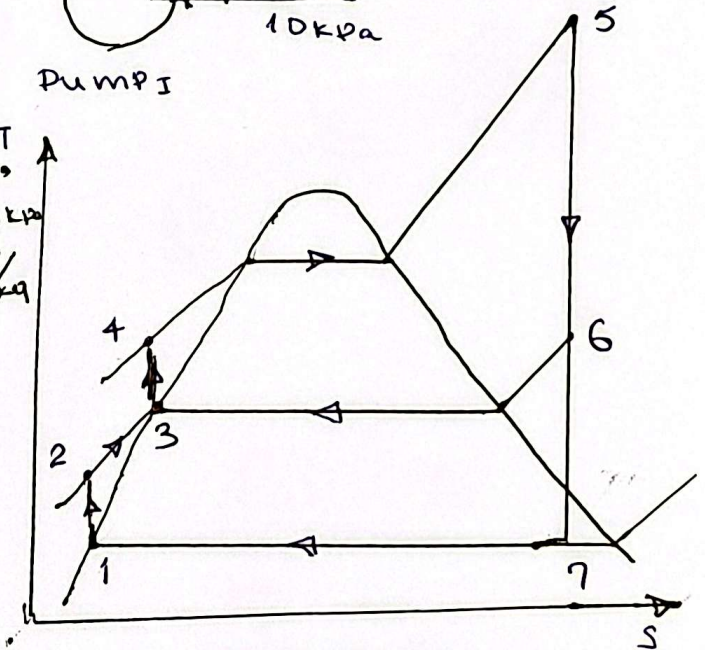
$$\begin{aligned} w_{P,I} &= v_1 (P_2 - P_1) \\ &= 0.00101 (1200 - 10) \text{ kJ/kg} \\ &= 1.2 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} h_2 &= h_1 + w_{P,I} \\ &= (191.81 + 1.21) \text{ kJ/kg} = 193.01 \text{ kJ/kg} \end{aligned}$$

$$\text{State 3: } P_3 = 1.2 \text{ MPa} \left\{ \begin{array}{l} v_3 = v_f @ 1.2 \text{ MPa} = 0.001138 \text{ m}^3/\text{kg} \\ h_3 = h_f @ 1.2 \text{ MPa} = 798.33 \text{ kJ/kg} \end{array} \right.$$

$$\text{State 4: } P_4 = 15 \text{ MPa} \\ s_4 = s_3$$

$$\begin{aligned} w_{P,II} &= v_3 (P_4 - P_3) \\ &= (0.001138) (15,000 - 1200) \text{ kPa} \\ &= 15.7 \text{ kJ/kg} \end{aligned}$$



-9-

$$h_4 = h_3 + W_{PI, in} = (798.33 + 15.70) \text{ kJ/kg} = 814.03 \text{ kJ/kg}$$
$$\text{State 5: } \left. \begin{array}{l} P_5 = 15 \text{ MPa} \\ T_5 = 600^\circ\text{C} \end{array} \right\} \begin{array}{l} h_5 = 3583.1 \text{ kJ/kg} \\ s_5 = 6.6796 \text{ kJ/kg}\cdot\text{K} \end{array}$$

$$\text{State 6: } \left. \begin{array}{l} P_6 = 1.2 \text{ MPa} \\ s_6 = s_5 \end{array} \right\} \begin{array}{l} h_6 = 2860.2 \text{ kJ/kg} \\ (T_6 = 218^\circ\text{C}) \end{array}$$

$$\text{State 7: } P_7 = 10 \text{ kPa}$$

$$x_7 = \frac{s_7 - s_f}{s_{fg}} = \frac{6.6796 - 0.6492}{7.4996} = 0.8041$$

$$h_7 = h_f + x_7 h_{fg} = 191.81 + 0.8041(2392.1) = 2115.3 \text{ kJ/kg}$$

Energy analysis of the mixing members

$$\dot{E}_{in} = \dot{E}_{out} \rightarrow \sum \dot{m} h_{in} = \sum \dot{m} h_{out}$$

or

$$y h_6 = (1-y) h_6 = 1(h_3)$$

where y is the fraction of steam extracted from the turbine ($= \dot{m}/\dot{m}_5$). Solving for y & substituting the enthalpy values, we find

$$y = \frac{h_3 - h_2}{h_6 - h_2} = \frac{798.33 - 193.01}{2860.2 - 193.01} = 0.2270$$

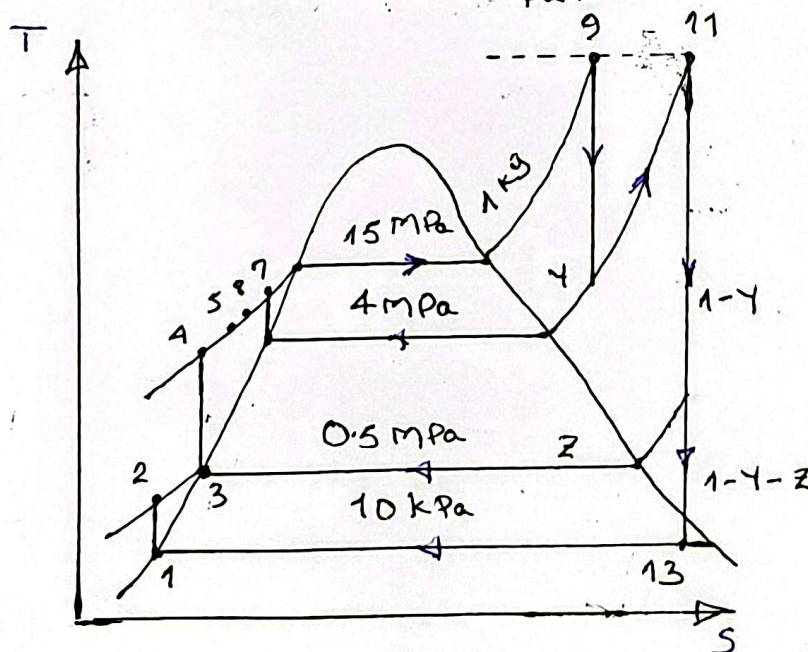
Thus,

$$\dot{q}_{in} = h_3 - h_4 = (3583.1 - 814.03) \text{ kJ/kg} = 2769.1 \text{ kJ/kg}$$

$$\dot{q}_{out} = (1-y)(h_7 - h_1) = (1-0.2270)(2115.3 - 191.81) \text{ kJ/kg} \\ = 1486.9 \text{ kJ/kg}$$

&

$$\eta_{th} = 1 - \frac{\dot{q}_{out}}{\dot{q}_{in}} = 1 - \frac{1486.9 \text{ kJ/kg}}{2769.1 \text{ kJ/kg}} = 0.43 \text{ or } 43\%$$



The enthalpies at the various state are:

$$h_1 = 191.81 \text{ kJ/kg}$$

$$h_9 = 3583.1 \text{ kJ/kg}$$

$$h_2 = 192.30 \text{ kJ/kg}$$

$$h_{10} = 3155.0 \text{ kJ/kg}$$

$$h_3 = 640.09 \text{ kJ/kg}$$

$$h_{11} = 3674.9 \text{ kJ/kg}$$

$$h_4 = 643.92 \text{ kJ/kg}$$

$$h_{12} = 3014.8 \text{ kJ/kg}$$

$$h_5 = 1087.4 \text{ kJ/kg}$$

$$h_{13} = 2335.7 \text{ kJ/kg}$$

$$h_6 = 1087.4 \text{ kJ/kg}$$

$$w_{pI, in} = 0.49 \text{ kJ/kg}$$

$$h_7 = 1101.2 \text{ kJ/kg}$$

$$w_{pII, in} = 3.83 \text{ kJ/kg}$$

$$h_8 = 1089.8 \text{ kJ/kg}$$

$$w_{pIII, in} = 13.77 \text{ kJ/kg}$$

The fractions of steam extracted are determined from the mass & energy balances of the feedwater heaters.

Closed feedwater heater:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$y h_{10} + (1-y) h_4 = (1-y) h_5 + y h_6$$

$$y = \frac{h_5 - h_6}{(h_{10} - h_6) + (h_5 - h_4)} = \frac{1087.4 - 643.92}{(3155 - 1087.4) + (1087.4 - 643.92)} = 0.1766$$

Open feed water heater:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$z h_{12} + (1-y-z) h_2 = (1-y) h_3$$

$$z = \frac{(1-y)(h_3 - h_2)}{h_{12} - h_2} = \frac{(1-0.1766)(640.09 - 192.30)}{3014.8 - 192.30} = 0.1306$$

The enthalpy at 8 is determined from by applying the mass & energy equations to the mixing chamber, which is assumed to be insulated:

$$\dot{E}_{in} = \dot{E}_{out}$$

$$(1) h_8 = (1-y) h_5 + y h_7$$

$$h_8 = (1-0.1766)(1087.4) \text{ kJ/kg} + 0.1766(1101.2) \text{ kJ/kg} = 1089.8 \text{ kJ/kg}$$

Thus,

$$\begin{aligned} \dot{q}_{in} &= (\dot{m}_9 - \dot{m}_8) + (1 - y)(\dot{m}_{11} - \dot{m}_{10}) \\ &= (3583.1 - 1089.8) \text{ kJ/kg} + (1 - 0.1766)(3674.9 - 3155) \text{ kJ/kg} \\ &= 2921.4 \text{ kJ/kg} \end{aligned}$$

$$\begin{aligned} \dot{q}_{out} &= (1 - y - z)(\dot{m}_{13} - \dot{m}_1) \\ &= (1 - 0.1766 - 0.1306)(2335.7 - 191.81) \text{ kJ/kg} \\ &= 1485.3 \text{ kJ/kg} \end{aligned}$$

∴

$$\eta_{th} = 1 - \frac{\dot{q}_{out}}{\dot{q}_{in}} = 1 - \frac{1485.3 \text{ kJ/kg}}{2921.4 \text{ kJ/kg}} = \underline{0.492 \text{ or } 49.2\%}$$